

## E15 Calculating Xseg and Xmm from RTD data

A sample of the tracer hythane was injected as a pulse into a reactor and the exit concentration was measured as a function of time. The data is given below.

1. Construct a curve showing  $C(t)$ .
2. Construct the  $E(t)$  curve.
3. Construct the  $F(t)$  curve.
4. Compute  $X_{CSTR}$ ,  $X_{PFR}$ ,  $X_{seg}$ , and  $X_{mm}$  for the following elementary liquid phase reactions:
  - a.  $A \rightarrow B$  with  $k = 0.1 \text{ min}^{-1}$
  - b.  $2A \rightarrow B$  with  $k = 0.1 \text{ m}^3/\text{kmol}/\text{min}$  and  $C_{A0} = 1.0 \text{ kmol}/\text{m}^3$

|                       |   |     |     |   |   |    |   |   |   |   |     |     |     |    |
|-----------------------|---|-----|-----|---|---|----|---|---|---|---|-----|-----|-----|----|
| t (min)               | 0 | 0.5 | 1   | 2 | 3 | 4  | 5 | 6 | 7 | 8 | 9   | 10  | 12  | 14 |
| C (g/m <sup>3</sup> ) | 0 | 0.6 | 1.4 | 5 | 8 | 10 | 8 | 6 | 4 | 3 | 2.2 | 1.6 | 0.6 | 0  |

consider  $2A \rightarrow B$   $-r_A = k C_A^2 = k C_{A0}^2 (1-x)^2$

CSTR  $V = \frac{F_{A0} X}{-r_A}$

$$V = \frac{F_{A0} X}{-r_A} = \frac{F_{A0} X}{k C_{A0}^2 (1-x)^2} = \frac{C_{A0} v_0 X}{k C_{A0}^2 (1-x)^2} = \frac{v_0 X}{k C_{A0} (1-x)^2}$$

$$\gamma = \frac{X}{k C_{A0} (1-x)^2}, \quad \gamma k C_{A0} = \frac{X}{(1-x)^2}$$

$\uparrow$   $\uparrow$   $\uparrow$   
 $t_m$   $0.1 \frac{\text{m}^3}{\text{kmol} \cdot \text{min}}$   $= 1 \frac{\text{kmol}}{\text{m}^3}$

$= 5.11 \text{ min}$

$$X_{\text{CSTR}} = 0.27$$

PFR:  $\frac{dX}{dV} = \frac{-r_A}{F_{A0}} = \frac{k C_{A0}^2 (1-X)^2}{F_{A0}} = \frac{k C_{A0}^2 (1-X)^2}{C_{A0} v_0}$

$$\frac{dX}{dV} = \frac{k C_{A0} (1-X)^2}{v_0}$$

$$\frac{dX}{d\tau} = k C_{A0} (1-X)^2 \quad \int_0^X \frac{dX}{(1-X)^2} = k C_{A0} \tau$$

$0.1 \frac{\text{kmol}}{\text{m}^3 \cdot \text{min}}$   
 $\downarrow$   
 $611 \text{ min}$   
 $\uparrow$   
 $1 \frac{\text{kmol}}{\text{m}^3}$

$$k C_{A0} \tau = \frac{X}{1-X}$$

$$X_{\text{PFR}} = 0.34$$

Segregation model  $\bar{X} = \int_0^\infty X(t) E(t) dt$

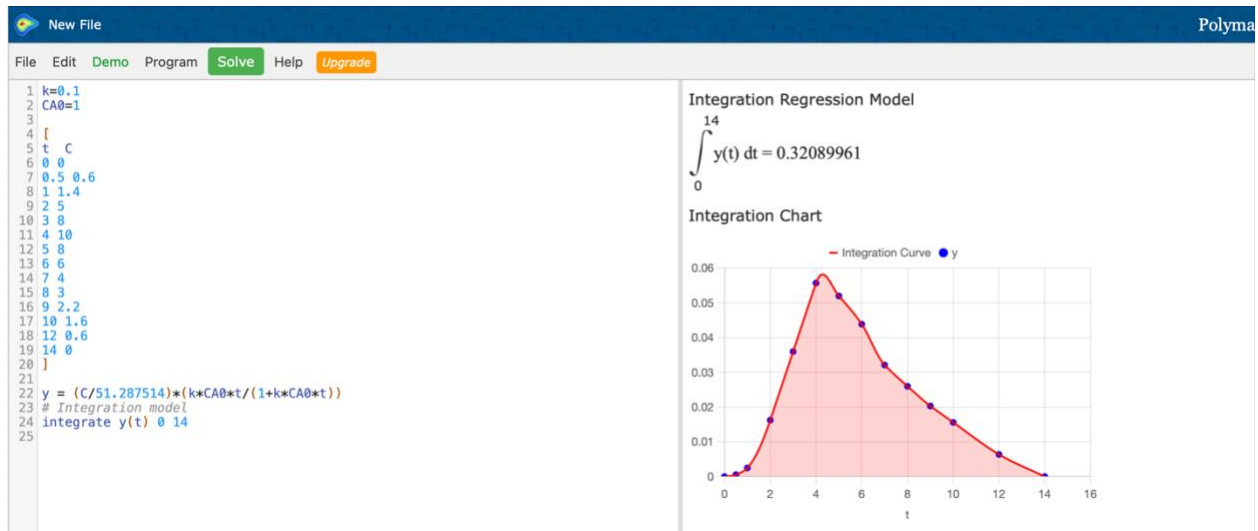
comes from the batch reactor mole balance

to get  $X(t)$ : write batch reactor mole balance

$$\frac{dx}{dt} = \frac{-r_A}{C_{A0}} = \frac{k C_{A0}^2 (1-x)^2}{C_{A0}} = k C_{A0} (1-x)^2$$

$$\int_0^x \frac{dx}{(1-x)^2} = k C_{A0} t \rightarrow \frac{x}{1-x} = k C_{A0} t$$

$$\text{Solve for } x: \quad x = \frac{k C_{A0} t}{1 + k C_{A0} t} = x(t)$$



$$\int_0^{\infty} X(t) E(t) dt = 0.32, \text{ determined w/ Polymath Plus}$$

$$X_{\text{seq}} = 0.32$$

maximum mixedness model

$$\frac{dX}{dz} = \frac{-r_A}{C_{A0}} - \frac{E(\bar{T}-z)}{1-F(\bar{T}-z)} X$$

$$z = t \quad \bar{T} = 14 \text{ min}$$

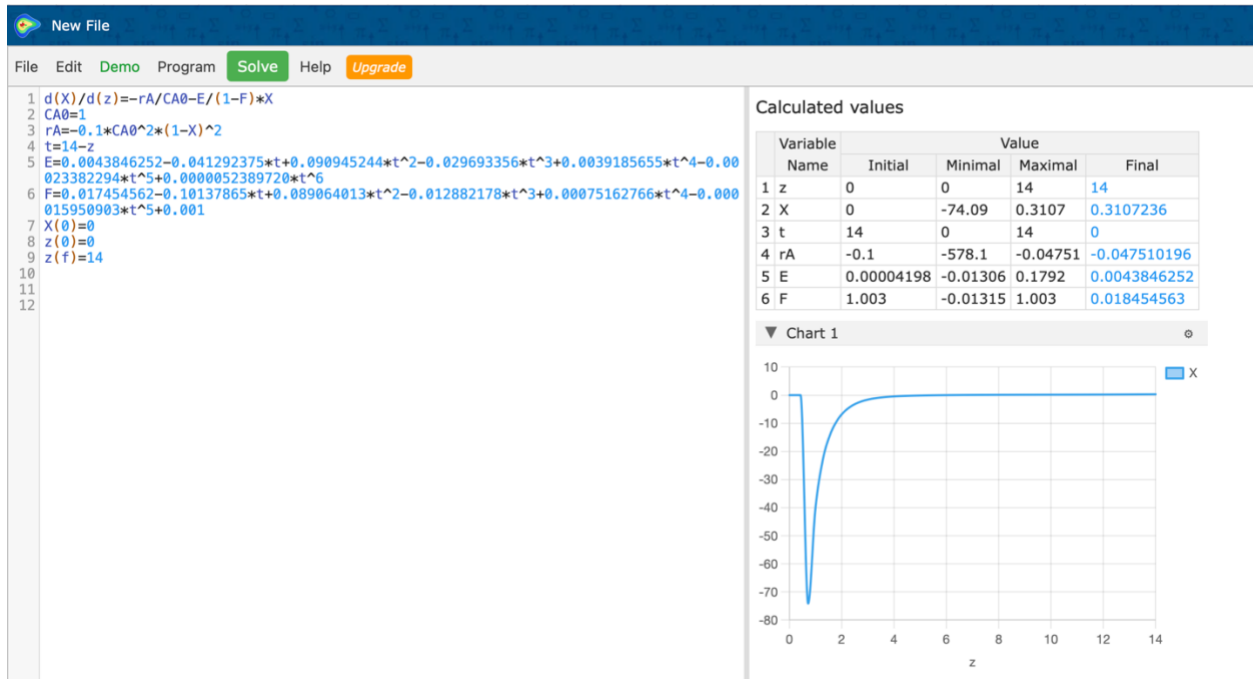
$$-r_A = k C_{A0}^2 (1-X)^2$$

$$C_{A0} = 1 \text{ kmol/m}^3$$

I am going to create best fit functions for  $E(t)$

&  $F(t)$  for the sake of ease

$$X_{mm} = 0.31$$



polynomials fit & data integrated using  
Polymath Plus

### Summary:

| <u>model</u> | <u>X</u> |
|--------------|----------|
| CSTR         | 0.27     |
| PFR          | 0.34     |
|              | 0.32     |
| Segregation  | 0.32     |
| mm           | 0.31     |